

3ART12 TOUCH

Variant: CHECKED

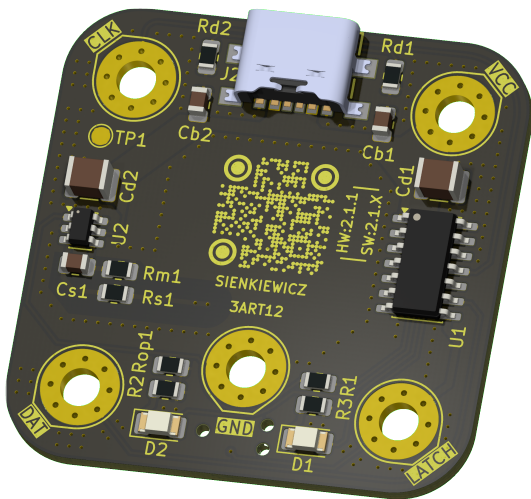
2026-06-08

Rev 2.1.1+ (Unreleased)

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TOP VIEW



BOTTOM VIEW



NOTES

Comment

Not fitted components are marked as X

- DRAFT - Very early stage of schematic, ignore details.
- PRELIMINARY - Close to final schematic.
- CHECKED - There shouldn't be any mistakes. Contact the engineer if you find any.
- RELEASED - A board with this schematic has been sent to production.

DESIGN CONSIDERATIONS

DESIGN NOTE:

Example text for informational design notes.

DESIGN NOTE:

Example text for debug notes.

DESIGN NOTE:

Example text for cautionary design notes.

DESIGN NOTE:

Example text for critical design notes.

LAYOUT NOTE:

Example text for critical layout guidelines.

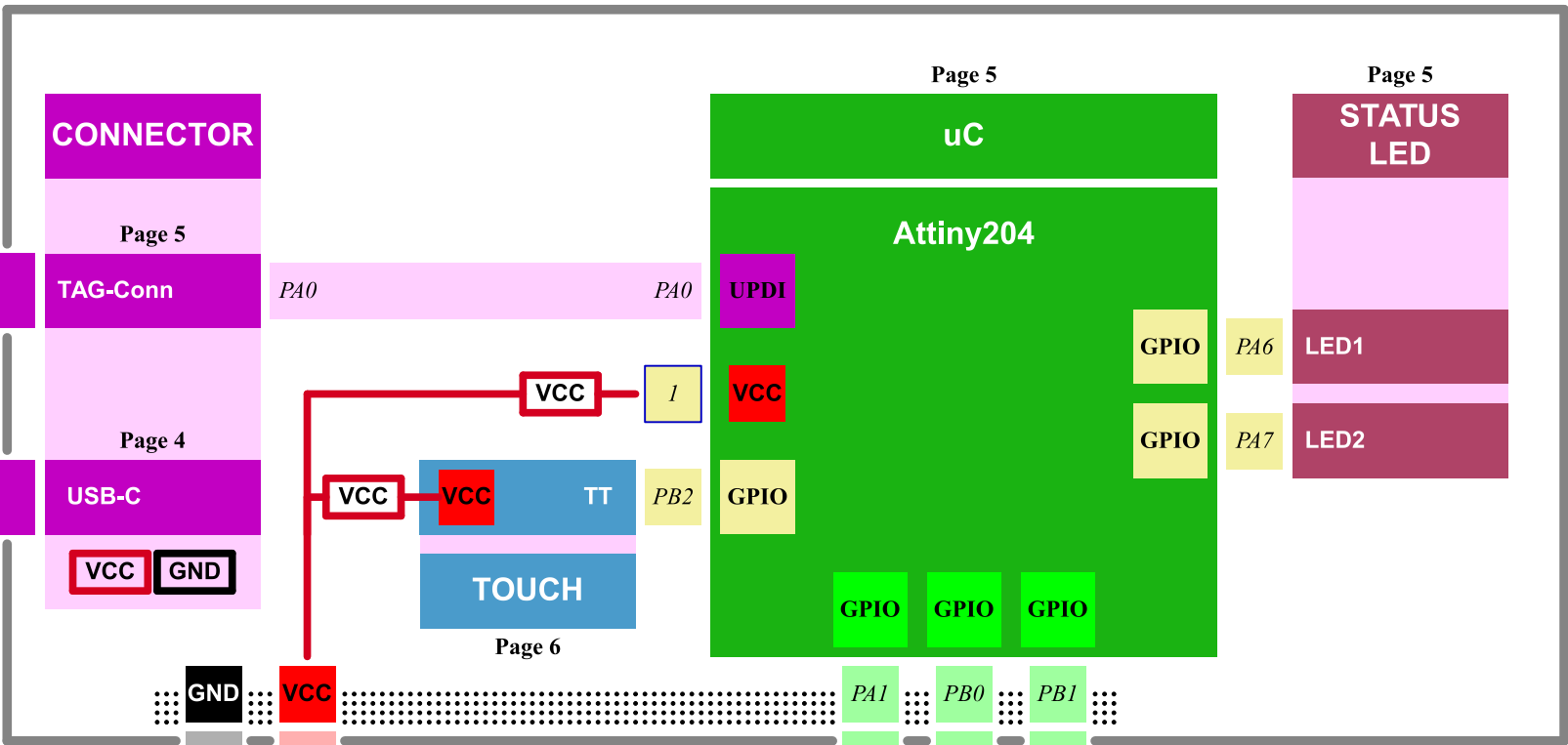
|  |                  |                                           |                              |                                  |
|--|------------------|-------------------------------------------|------------------------------|----------------------------------|
|  | Comments:        | Company:<br>BaJRan imaginary industries   | Variant:<br>CHECKED          | Git Hash:<br>5ba43f3             |
|  | Sheet Title:     | Board Name:<br>3ART12 TOUCH               | Project Name:<br>Sienkiewicz |                                  |
|  | Sheet Path:<br>/ | File Name:<br>Sienkiewicz_TOUCH.kicad_sch | Designer:<br>Pawel Baran     | Date:<br>2025-01-12              |
|  |                  |                                           | Reviewer:                    | Revision:<br>2.1.1+ (Unreleased) |
|  |                  |                                           | Size:<br>A3                  | Sheet:<br>1 of 9                 |

[2] Block Diagram

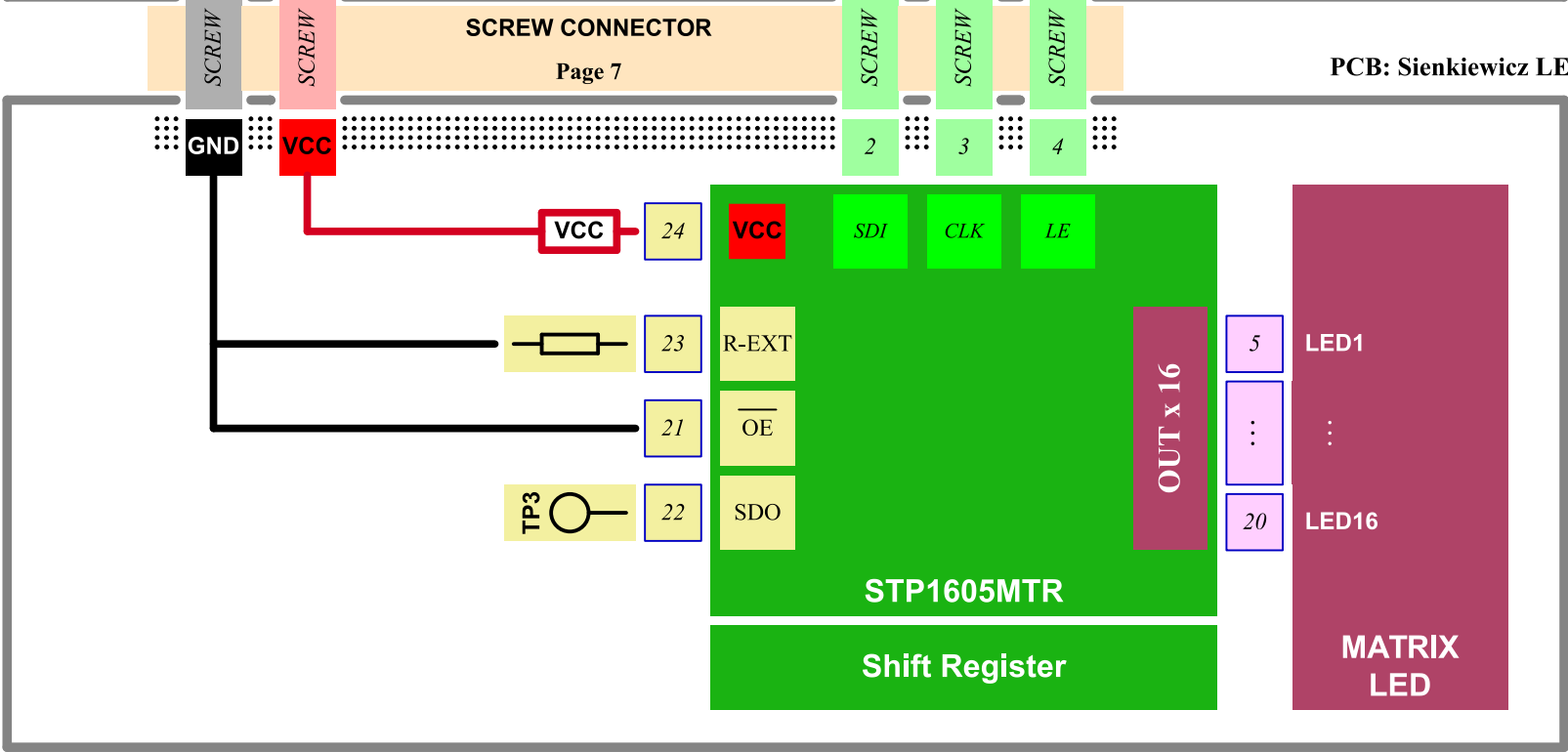
PCB: Sienkiewicz TOUCH

Programmer

USB-C Cable



PCB: Sienkiewicz LED

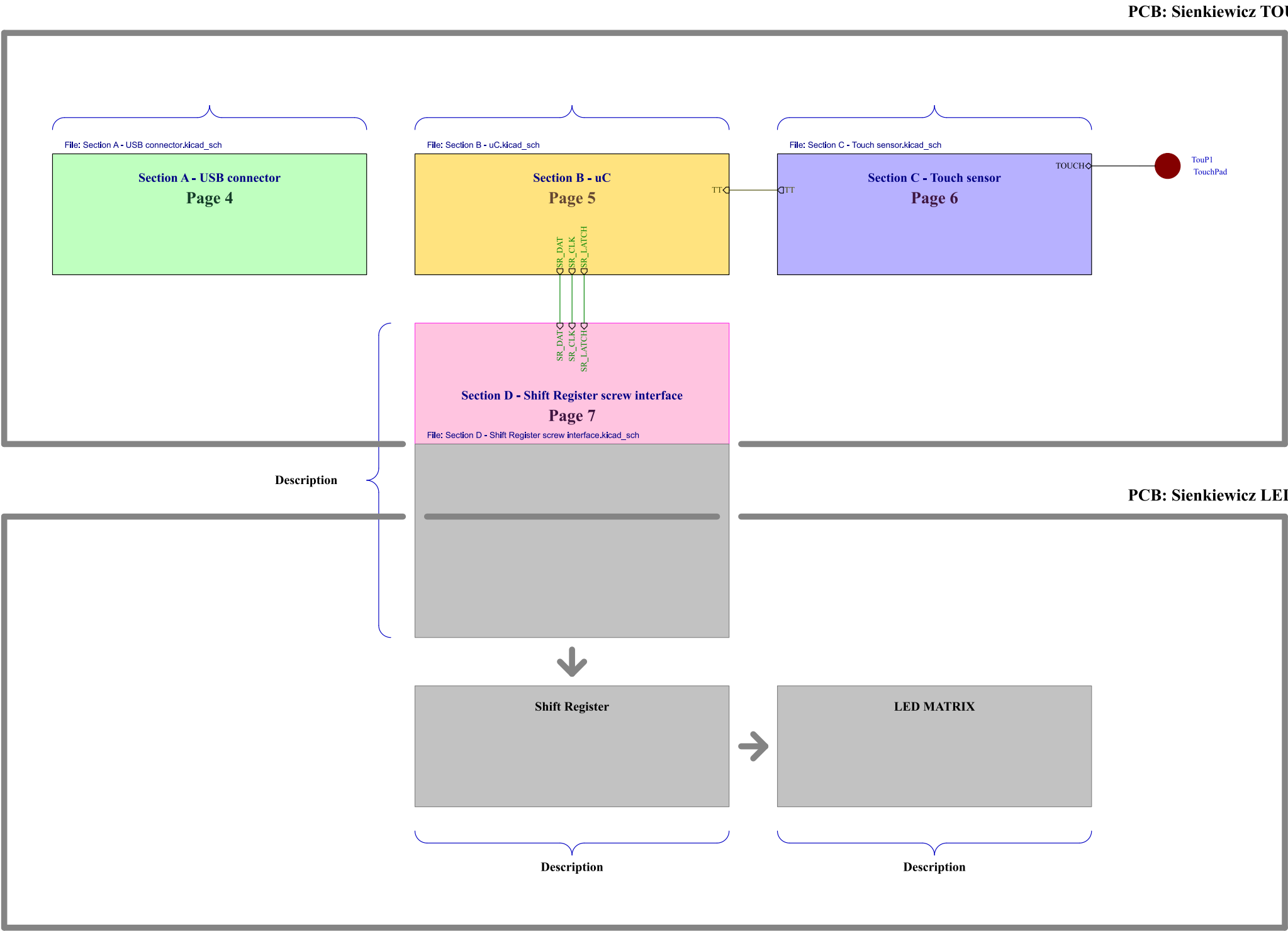


Target specifications:

|               |       |
|---------------|-------|
| Input voltage | 5V    |
| Current       | 200mA |
| Spec 3        | ----  |
| Spec 4        | ----  |

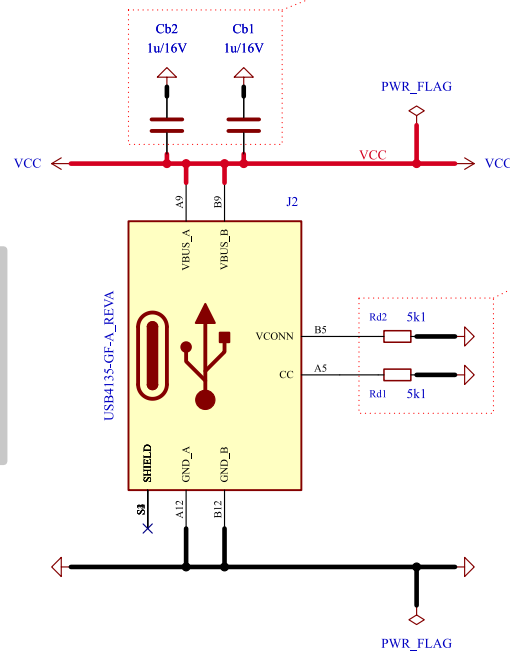
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|                                | Board Name:<br>3ART12 TOUCH              |                                       | Project Name:<br>Sienkiewicz |                                  |
|                                | Sheet Title:<br>Block Diagram            | File Name:<br>Block Diagram.kicad_sch | Designer:<br>Pawel Baran     | Date:<br>2025-01-12              |
| Sheet Path:<br>/Block Diagram/ |                                          |                                       | Reviewer:                    | Size:<br>A3                      |
|                                |                                          |                                       |                              | Revision:<br>2.1.1+ (Unreleased) |
|                                |                                          |                                       |                              | Sheet:<br>2 of 9                 |

# [3] Project Architecture



|                                       |                                      |                                              |                                          |                     |                                  |                      |
|---------------------------------------|--------------------------------------|----------------------------------------------|------------------------------------------|---------------------|----------------------------------|----------------------|
|                                       | Comments:                            |                                              | Company:<br>BaJRan imgaginary industries |                     | Variant:<br>CHECKED              | Git Hash:<br>5ba43f3 |
|                                       | Board Name:<br>3ART12 TOUCH          |                                              | Project Name:<br>Sienkiewicz             |                     |                                  |                      |
|                                       | Sheet Title:<br>Project Architecture | File Name:<br>Project Architecture.kicad_sch | Designer:<br>Pawel Baran                 | Date:<br>2025-01-12 | Revision:<br>2.1.1+ (Unreleased) |                      |
| Sheet Path:<br>/Project Architecture/ |                                      |                                              | Reviewer:                                | Size:<br>A3         | Sheet:<br>3 of 9                 |                      |

## [4] USB connector



DESIGN NOTE:  
Cb1, Cb2  
bulk capacitor

Release 2.0  
August 2019

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USB Type-C Cable and  
Connector Specification

Table 4-2 VBUS Source Characteristics

|                        | Minimum | Maximum | Notes                                                                                                     |
|------------------------|---------|---------|-----------------------------------------------------------------------------------------------------------|
| VBUS Leakage Impedance | 72.4 kΩ |         | Leakage between VBUS pins and GND pins on receptacle when VBUS is not being sourced.                      |
| Vbus Capacitance       |         | 3000 μF | Capacitance for source-only ports between VBUS and GND pins on receptacle when Vbus is not being sourced. |
|                        |         | 10 μF   | Capacitance for DRP ports between VBUS and GND pins on receptacle when Vbus is not being sourced.         |

DESIGN NOTE:

Per USB-IF specification, a sink role is established by placing 5.1 kΩ pull-down (Rd) resistors on CC1 and CC2.

Release 2.0  
August 2019]

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USB Type-C Cable and  
Connector Specification

Table 4-25 Sink CC Termination (Rd) Requirements

| Rd Implementation                | Nominal value | Can detect power capability? | Max voltage on pin |
|----------------------------------|---------------|------------------------------|--------------------|
| ± 20% voltage clamp <sup>1</sup> | 1.1 V         | No                           | 1.32 V             |
| ± 20% resistor to GND            | 5.1 kΩ        | No                           | 2.18 V             |
| ± 10% resistor to GND            | 5.1 kΩ        | Yes                          | 2.04 V             |

Notes:

1. The clamp implementation inhibits *USB PD* communication although the system can start with the clamp and transition to the resistor once it is able to do *USB PD*.

[www.usb.org/sites/default/files/USB%20Type-C%20Spec%20R2.0%20-%20August%202019.pdf](http://www.usb.org/sites/default/files/USB%20Type-C%20Spec%20R2.0%20-%20August%202019.pdf)

Comments:

AN5346  
STM32G474 Datasheet p.81  
J. Pieper ADC investigation

Company:

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Variant:

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Board Name:

3ART12 TOUCH

Project Name:

Sienkiewicz

Sheet Title:

USB connector

File Name:

Section A - USB connector.kicad\_sch

Designer:

Pawel Baran

Date:

2025-01-12

Revision:

2.1.1+ (Unreleased)

Sheet Path:

/Project Architecture/Section A - USB connector/

Reviewer:

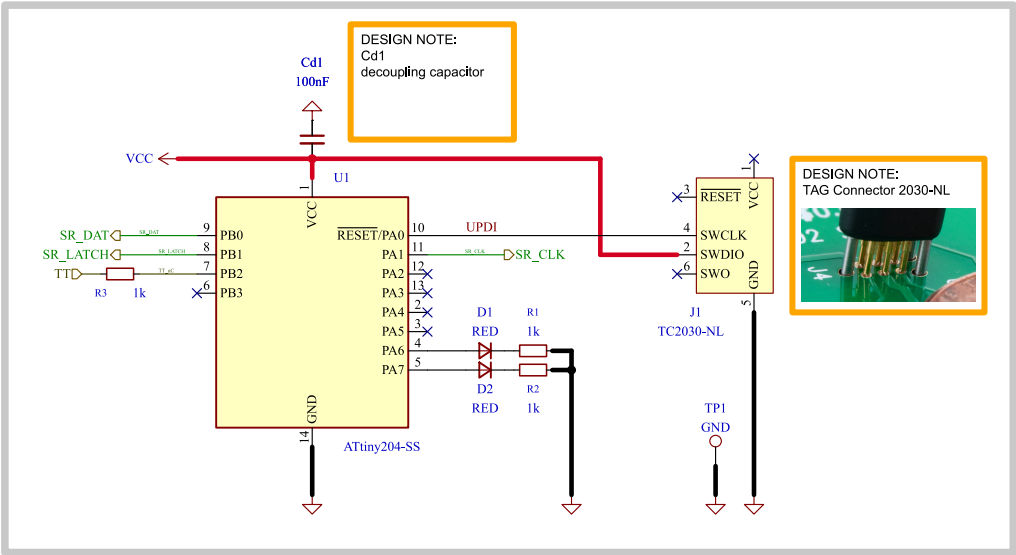
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A4

Sheet:

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# [5] uC & prog connector



|  |                                                      |                                          |                          |                              |                                  |
|--|------------------------------------------------------|------------------------------------------|--------------------------|------------------------------|----------------------------------|
|  | Comments:                                            | Company:<br>BaJRan imgaginary industries |                          | Variant:<br>CHECKED          | Git Hash:<br>5ba43f3             |
|  |                                                      | Board Name:<br>3ART12 TOUCH              |                          | Project Name:<br>Sienkiewicz |                                  |
|  | Sheet Title:<br>uC & prog connector                  | File Name:<br>Section B - uC.kicad_sch   | Designer:<br>Pawel Baran | Date:<br>2026-04-16          | Revision:<br>2.1.1+ (Unreleased) |
|  | Sheet Path:<br>/Project Architecture/Section B - uC/ |                                          | Reviewer:                | Size:<br>A4                  | Sheet:<br>5 of 9                 |

# [6] Touch Sensor

## DESIGN NOTE:

Table 3-2. Predefined Auto-off Delay (Active High Output)

| Vt  | Auto-off Delay (t <sub>o</sub> )          |
|-----|-------------------------------------------|
| Vss | Infinity (remain on until toggled to off) |
| Vdd | 15 minutes                                |
| OUT | 60 minutes                                |

Table 3-3. Predefined Auto-off Delay (Active Low Output)

| Vt  | Auto-off Delay (t <sub>o</sub> )          |
|-----|-------------------------------------------|
| Vss | 15 minutes                                |
| Vdd | Infinity (remain on until toggled to off) |
| OUT | 60 minutes                                |

## DESIGN NOTE:

### Output Polarity Selection

The output (OUT pin) of the QT1012 can be configured to have an active high or active low output by means of the output configuration resistor R<sub>op</sub>. The resistor is connected between the output and either Vss or Vdd (see Figure 3-4 and Table 3-1). A typical value for R<sub>op</sub> is 100 kΩ.

Table 3-1. Output Configuration

| Name (V <sub>op</sub> ) | Function (Output Polarity) |
|-------------------------|----------------------------|
| Vss                     | Active high                |
| Vdd                     | Active low                 |

DESIGN NOTE:  
Cd2  
decoupling  
capacitor

DESIGN NOTE:  
sense circuit  
Cs [2nF - 60nF]  
Rs [4k7 - 20k]

## DESIGN NOTE:

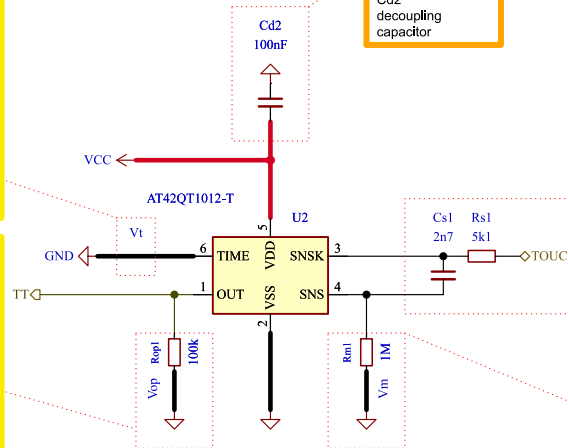
A single 1 MΩ resistor (R<sub>m</sub>) is connected between the SNS pin and the logic level V<sub>m</sub> to provide three auto-off functions: delay multiplication, delay override and delay retriggering. On power-up the logic level at V<sub>m</sub> is assessed and the delay multiplication factor is set to x1 or x24 accordingly (see Figure 3-6 on page 12, Table 3-2 on page 12 and Table 3-3 on page 12). At the end of each acquisition cycle the logic level of V<sub>m</sub> is monitored to see if an Auto-off delay override is required (see Section 3.11.5 on page 17). Setting the delay multiplier to x24 will decrease the key sensitivity. To compensate, it may be necessary to increase the value of Cs.

Table 3-4. Auto-off Delay Multiplier

| V <sub>m</sub> | Auto-off Delay Multiplier |
|----------------|---------------------------|
| Vss            | t <sub>o</sub> × 1        |
| Vdd            | t <sub>o</sub> × 24       |

ACTIVE HIGH OUTPUT

AUTO DELAY OFF



Comments:

Company:

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Variant:

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Git Hash:

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Board Name:

3ART12 TOUCH

Project Name:

Sienkiewicz

Sheet Title:

Touch Sensor

File Name:

Section C - Touch sensor.kicad\_sch

Designer:

Pawel Baran

Date:

2026-04-14

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/Project Architecture/Section C - Touch sensor/

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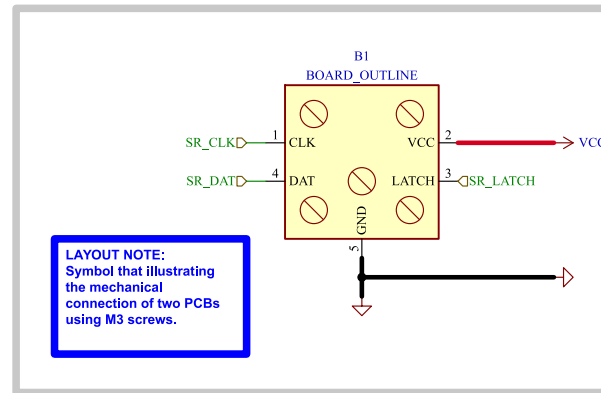
Size:

A4

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# [7] Shift Register screw interface



|  |                                                                   |                                                      |             |               |           |
|--|-------------------------------------------------------------------|------------------------------------------------------|-------------|---------------|-----------|
|  | Comments:                                                         | Company:                                             |             | Variant:      | Git Hash: |
|  |                                                                   |                                                      |             | CHECKED       | 5ba43f3   |
|  |                                                                   | Board Name:                                          |             | Project Name: |           |
|  |                                                                   | 3ART12 TOUCH                                         |             | Sienkiewicz   |           |
|  | Sheet Title:                                                      | File Name:                                           | Designer:   | Date:         | Revision: |
|  | Shift Register screw interface                                    | Section D - Shift Register screw interface.kicad_sch | Pawel Baran | 2026-04-16    |           |
|  | Sheet Path:                                                       | Reviewer:                                            | Size:       | Sheet:        |           |
|  | /Project Architecture/Section D - Shift Register screw interface/ |                                                      | A4          | 7 of 9        |           |

# [8] Requirements

|                |                                                                                                                                                                                                                                                           |      |                                                                                                                                             |
|----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|---------------------------------------------------------------------------------------------------------------------------------------------|
| Project        | Simplified electronic project designed to verify practical skills in electrical engineering, PCB design, electronic assembly, embedded software implementation, and programming within the Microchip Technology development environment. Simple powering. | MEET | The project is a simple LED light effect device with a user interface realized using a capacitive touch sensor. Powered by USB-C Connector. |
| MAIN OBJECTIVE |                                                                                                                                                                                                                                                           |      |                                                                                                                                             |

|    |                                                                                            |      |                                                                                                                              |
|----|--------------------------------------------------------------------------------------------|------|------------------------------------------------------------------------------------------------------------------------------|
| 1  | The project shall contain very small integrated circuits.<br>Pitch less than 1mm           | MEET | IC: Cap sensor : AT42QT1012 : SOT-23-6 package : PITCH: 0.95<br>IC: uC : Attiny-204 : SOIC-14_3.9x8.7mm_P1.27mm : PITCH 1.27 |
| 2  | The project shall contain 0805 SMT passives components.                                    | MEET | Device contain multiple 0805 SMT elements.                                                                                   |
| 3  | The project shall contain multi-pin integrated circuits to verify soldering precision.     | MEET | IC: Shift register : STP16CP05MTR : SO-24 : 24 pins<br>IC: uC : Attiny204 : SOIC-14_3.9x8.7mm : 14 pins                      |
| 4  | The project shall contain an UPDI programming interface for firmware upload and debugging. | MEET | uC Attiny204 with UPDI programming interface                                                                                 |
| 5  | Possibility to program using Microchip enviroment and pickit-4 programmer.                 | MEET | It is possible to program the device using pickit-4 and MPLAB software.                                                      |
| 6  | Shall contain at least two status led.                                                     | MEET | Status LED'S : LED1, LED2.                                                                                                   |
| 7  | Shall be able to interact with user.                                                       | MEET | The user can change the light effect using a capacitive touch sensor.                                                        |
| 8  | Shall be easy to power up.                                                                 | MEET | Powered by USB-C connector with general mode                                                                                 |
| 9  |                                                                                            |      |                                                                                                                              |
| 10 |                                                                                            |      |                                                                                                                              |
| 11 |                                                                                            |      |                                                                                                                              |
| 12 |                                                                                            |      |                                                                                                                              |

|  |  |                               |                                            |                          |                              |                                  |
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|  |  |                               | Board Name:<br>3ART12 TOUCH                |                          | Project Name:<br>Sienkiewicz |                                  |
|  |  | Sheet Title:<br>Requirements  | File Name:<br>Power - Sequencing.kicad_sch | Designer:<br>Pawel Baran | Date:<br>2025-01-12          | Revision:<br>2.1.1+ (Unreleased) |
|  |  | Sheet Path:<br>/Requirements/ |                                            | Reviewer:                | Size:<br>A4                  | Sheet:<br>8 of 9                 |



# [9] Revision History

## Version 2.0.0 - 2026-04-24

- Added
- First 2.0.0 Version of pcb and schematic

## Version 2.0.1 - 2026-04-25

- Changed
- Modified silkscreen look, by adding bigger QR-code. small rearangement.

## Version 2.0.2 - 2026-04-26

- Changed
- Modified silkscreen look
  - Fabrication layers update

## Version 2.1.0 - 2026-05-01

- Changed
- B-B connector maping changes, CLK and SDI swapped (fixes #4)

## Version 2.1.1 - 2026-05-21

- Changed
- B-B connector maping changes restored (fixes #4)

- Added
- requirements section added for documentation

|  |                                   |                                          |                          |                              |                                  |
|--|-----------------------------------|------------------------------------------|--------------------------|------------------------------|----------------------------------|
|  | Comments:                         | Company:<br>BaJRan imgaginary industries |                          | Variant:<br>CHECKED          | Git Hash:<br>5ba43f3             |
|  |                                   | Board Name:<br>3ART12 TOUCH              |                          | Project Name:<br>Sienkiewicz |                                  |
|  | Sheet Title:<br>Revision History  | File Name:<br>Revision History.kicad_sch | Designer:<br>Pawel Baran | Date:<br>2025-01-12          | Revision:<br>2.1.1+ (Unreleased) |
|  | Sheet Path:<br>/Revision History/ |                                          | Reviewer:                | Size:<br>A4                  | Sheet:<br>9 of 9                 |